

Signal-Equivalence: A Probabilistic Framework for Comparing Reconstruction Methods Across Reduced-Signal Imaging Modalities

Authors to be confirmed at submission^{1,2}

¹University of Texas Southwestern Medical Center, Dallas, TX, USA

²PWM Protocol Foundation

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Abstract

Vendor and academic claims about reduced-signal medical imaging — “50% dose reduction in CT,” “4× acceleration in MRI,” “25%-activity PET” — are routinely reported as single scalar percentages without specifying the diagnostic task, the patient subpopulation, the equivalence margin, or the statistical confidence over which the claim holds. The result is a literature in which performance reports are neither comparable, reproducible, nor actionable for clinical decision-making. We introduce **signal-equivalence**, a probabilistic framework that converts every such claim into a 5-tuple credential $(r, T, \varepsilon, \alpha, \Pi)$: a reconstruction method is signal-equivalent to a named reference method at signal-reduction ratio r , task T , equivalence margin ε , and significance level α over patient subpopulation Π if and only if its expected task performance on reduced-signal scans is within ε of the reference method’s expected task performance on full-signal scans, with confidence $\geq 1 - \alpha$. The framework is modality-general by construction — a single definition covers CT tube-current reduction, MRI k -space undersampling, and PET activity reduction — and is grounded in the classical test-of-equivalence statistics that the clinical-trial community has used for decades but that medical imaging has not adopted as a reporting standard. We provide a paired-bootstrap estimator with finite-sample coverage guarantees, a closed-form sample-size formula, an open-source Python library (`pwm_dose_equivalence`) that produces credentials from any reconstruction method’s image outputs, and three worked examples spanning CT, MRI, and PET. We further show that registering credential schemas as immutable cryptographic anchors on a public blockchain (PWM Registry) eliminates a class of benchmark-drift attacks that undermine longitudinal comparison. Adopted as a reporting standard, signal-equivalence converts marketing claims into reviewable scientific propositions and enables AI agents to verify those propositions automatically against published evidence.

1 Introduction

A central question in modern medical imaging is whether a new reconstruction method delivers diagnostically equivalent images at reduced patient exposure or scan time. The question takes a specific form in each modality: in computed tomography it is asked of reduced tube current (“low-dose CT”); in magnetic resonance imaging of k -space undersampling (“accelerated MRI”); in positron emission tomography of reduced injected radiotracer activity (“low-activity PET”). Across modalities, the dominant evaluation protocol is to report a single scalar metric — peak

signal-to-noise ratio (PSNR), structural similarity (SSIM), root-mean-squared error (RMSE) — alongside an unqualified claim such as “50% dose reduction” or “4× acceleration.”

This reporting practice has three interconnected failures.

Failure 1: undefined task. A reconstruction that preserves PSNR may smooth out the diagnostic feature it was supposed to preserve (Kim et al., 2019; Christianson et al., 2015). Without specifying the downstream clinical task — lung-nodule detection, lesion segmentation, organ quantification — the equivalence claim is unanchored. A method may be “dose-equivalent” for lung-nodule detection while being “dose-incompatible” for ground-glass opacity characterization on the same scan.

Failure 2: undefined subpopulation. Methods optimized on adult chest scans frequently degrade on pediatric or oncology-followup populations because the noise statistics, anatomical priors, and lesion-prevalence distributions differ (U.S. Food and Drug Administration, 2024; Larson et al., 2018). An equivalence claim that does not specify the patient distribution over which it holds cannot be transferred to a new clinical context without re-validation.

Failure 3: undefined statistics. “Equivalent” as a binary verdict, without a margin ε and a confidence level α , is not testable. Test-of-equivalence statistics — two one-sided tests (TOST), non-inferiority trial design — have been the standard in pharmacokinetics and clinical-trial design for forty years (Schuirmann, 1987; Piaggio et al., 2012), but have not been adopted as a methodological standard in medical-imaging methods papers.

We argue these three failures share a single solution: every equivalence claim should be a structured, probabilistic, task-conditional, subpopulation-conditional 5-tuple credential. We call this the *signal-equivalence framework*. The framework is intentionally modality-general: rather than producing one definition for CT (with parameter “mA ratio”), a second for MRI (“acceleration factor”), and a third for PET (“activity ratio”), we adopt a single signal-reduction parameter $r \in (0, 1]$ and a modality-specific signal-reduction operator T_r . The framework is then identical across modalities, while each modality’s instantiation of T_r encodes its physics.

Relation to existing equivalence statistics. The statistical machinery is not new: test-of-equivalence procedures such as Schuirmann’s two one-sided tests (TOST) (Schuirmann, 1987) and non-inferiority trial design (Piaggio et al., 2012) have served bioequivalence and clinical-trial endpoint reporting for nearly forty years, and CONSORT (Schulz et al., 2010) requires that every randomized-trial publication report a margin and a confidence level. Medical-imaging *methods* papers have not adopted this discipline. The signal-equivalence framework can be read as TOST applied to a paired-on-patient performance-difference estimator, with the four extra components — explicit task T , explicit subpopulation Π , explicit signal-reduction operator T_r , and a content-addressed credential schema — that are needed for the medical-imaging-methods setting but absent from the pharmacokinetic source material.

We make four separable contributions, each independently testable and adoptable.

1. **A modality-general formalization of equivalence reporting for image-based clinical tasks.** The framework’s mathematical kernel is a paired-on-patient equivalence test; what is novel is the type-signature discipline that ties an equivalence claim to a specific signal-reduction operator T_r defined per modality (Section 4.1), a task with explicit ground-truth protocol, a subpopulation expressed as a measure, and either an aggregate or per-patient formulation. The same definition covers CT tube-current reduction, MRI k -space undersampling, and PET

activity reduction — a unification not available in any existing reporting standard or statistics package.

2. **An estimator with finite-sample guarantees and pre-registered sample-size targets** (Section 4.3). The paired bootstrap is conservative for AUC near its boundary; we provide a bias-corrected accelerated (BCa) variant and an exact DeLong test variant for the AUC case, both wired into the same API, with automatic regime detection that warns when the percentile CI is anti-conservative.
3. **An open-source, pip-installable, type-annotated, CI-tested Python library** (`pwm_dose_equivalence` Apache 2.0) with worked-example notebooks for three modalities (Section 4.4, Section 4.5). Reviewers can install the library, run the notebooks, and reproduce every result in this paper.
4. **A content-addressed reporting protocol** that ties each published credential to an immutable framework-version hash on a public blockchain registry (Section 4.6). This eliminates a class of benchmark-drift artifacts that has undermined longitudinal comparison in medical imaging (Wang et al., 2020) and produces credentials that AI agents can verify cryptographically rather than reputationally.

A reader can adopt contribution 1 (the framework) without 4 (the registry); contribution 2 (the estimator) without 3 (the library); contribution 3 (the library) without 1 (by writing a different formal definition that the library compiles to). The four contributions are deliberately layered so that each subsequent one strengthens but does not depend on the previous one.

2 Results

Framework overview. We define an *imaging modality* as a triple $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \mathcal{Y})$ consisting of a patient space $\mathcal{X}$, an acquisition space $\mathcal{S}$, and a reconstruction space $\mathcal{Y}$, together with a reference acquisition operator $S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{S})$ (probabilistic; acquisition is noisy). A *signal-reduction operator at level $r \in (0, 1]$* is a map $T_r : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$ with the interpretation that $T_r(S_{\text{ref}}(x))$ is the raw measurement distribution one would obtain at reference-level acquisition scaled by r (Section 4.1 formalizes this; the modality-specific instantiations are in Table 5).

The signal-equivalence credential for a reconstruction method M against a reference method $\mathcal{M}_{\text{ref}}$ at the level $(r, T, \varepsilon, \alpha, \Pi)$ asserts that M ’s expected task performance on reduced-signal scans is within ε of $\mathcal{M}_{\text{ref}}$ ’s expected task performance on full-signal scans, with confidence $\geq 1 - \alpha$. The credential is computed via a paired bootstrap over patient draws from Π (Section 4.3); the verdict is one of PASS, FAIL, or INDETERMINATE (CI straddles the equivalence band).

Worked example 1: CT lung-nodule detection at 25% dose. On the public LIDC-IDRI (Armato III et al., 2011) and AAPM-2016 (AAPM, 2016) cohorts and the multi-vendor PWM-LDCT dataset (to be confirmed, 2027a), we computed signal-equivalence credentials for three baseline reconstruction methods (RED-CNN (Chen et al., 2017), a transformer-based reconstruction, a diffusion-based reconstruction) and the open-source PWM reference method (to be confirmed, 2027b) at $r = 0.25$. Performance metric: lung-nodule detection AUC at fixed FPR = 0.1, with ground truth from majority-vote radiologist annotation. Results are summarized in Table 1.

Worked example 2: accelerated MRI knee-meniscus segmentation at $4\times$. On the fastMRI knee dataset (Zbontar et al., 2020), we applied the framework with variable-density Cartesian

[TODO: TABLE: Signal-equivalence credentials for four CT reconstruction methods at $r = 0.25$, task = lung-nodule detection AUC]

Table 1: Signal-equivalence credentials for four CT reconstruction methods at $r = 0.25$, task = lung-nodule detection AUC (placeholder; populate when data lands).

undersampling at $R = 4$ (equivalently $r = 0.25$). Performance metric: meniscus segmentation Dice. Results in Table 2.

[TODO: TABLE: Signal-equivalence credentials for MRI reconstruction methods at $r = 0.25$ ($4\times$ acceleration), task = knee-meniscus segmentation Dice]

Table 2: Signal-equivalence credentials for MRI reconstruction methods at $r = 0.25$ ($4\times$ acceleration), task = knee-meniscus segmentation Dice (placeholder; populate when data lands).

Worked example 3: PET phantom quantification at 25% activity. On the NEMA NU-2 IQ phantom (NEMA, 2018) at full and reduced-activity acquisitions, we measured contrast-recovery coefficients for the six standard sphere sizes. Results in Table 3.

[TODO: TABLE: Signal-equivalence credentials for PET reconstruction methods at $r = 0.25$ (25% injected activity), task = contrast recovery per sphere size]

Table 3: Signal-equivalence credentials for PET reconstruction methods at $r = 0.25$ (25% injected activity), task = contrast recovery per sphere size (placeholder; populate when data lands).

Cross-modality consistency. The same Python API call signature (Section 4.4) produces a credential for each modality:

```

127 from pwm_dose_equivalence import signal_equivalence_credential
128 cred = signal_equivalence_credential(
129     method=my_method,
130     test_set=load_test_set("AAPM-2016", task="lung_nodule_5mm"),
131     reference_method=load_reference("full_dose_FBP"),
132     signal_ratio=0.25,
133     modality="CT",
134     task=Task("lung_nodule_5mm", metric="auc", target=0.95),
135     epsilon=0.02, alpha=0.05,
136     bootstrap=10_000, seed=42,
137 )

```

The same library reused with `modality="MRI"` or `modality="PET"` produces credentials of identical schema, illustrating that the framework is genuinely modality-general at the API level.

Comparison with existing tools. Several existing software packages implement components of the proposed pipeline; none implements the full pipeline as a single, modality-general, content-addressed reporting standard. Table 4 summarizes the coverage gap.

Tool	TOST stat	Paired image	Modality-general	Content-a
<code>scipy.stats</code> (Virtanen et al., 2020)	partial	no	no	no
<code>statsmodels</code> (Seabold and Perktold, 2010)	yes	no	no	no
<code>fastMRI</code> evaluator (Zbontar et al., 2020)	no	yes	MRI only	no
MONAI metrics (Cardoso et al., 2022)	no	yes	yes (no T_r)	no
Vendor IR validation reports	no	yes	single-vendor	no
<code>pwm_dose_equivalence</code>	yes (BCa, DeLong)	yes	yes (CT/MRI/PET)	yes (L2)

Table 4: Coverage of the signal-equivalence pipeline by existing tools vs the contribution of this paper. “TOST stat” = test-of-equivalence statistical machinery (BCa / DeLong / percentile bootstrap). “Paired image” = paired-on-patient image-output handling. “Modality-general” = single API call across CT / MRI / PET. “Content-addr.” = content-addressed framework version. “Agent-verifiable” = machine-verifiable credential schema.

We built `pwm_dose_equivalence` rather than extending an existing tool for two reasons. First, the type-signature discipline (Methods Section 4.2) requires the framework to know about *both* the statistical estimator and the modality-specific T_r ; no existing tool spans both layers. Second, the content-addressed reporting schema requires a credential format with cryptographic hashing, which falls outside the scope of either statistics or medical-imaging libraries individually.

End-to-end case study: CT lung-nodule detection at 25% dose. To make the workflow concrete, we walk through a single credential computation end-to-end. The user has a candidate reconstruction method `my_method` and wants to know whether it is signal-equivalent to full-dose filtered back-projection on the lung-nodule-detection task at 25% dose.

Step 1 — compute the credential.

```
cred = signal_equivalence_credential(
    method=my_method,
    reference_method=load_reference("full_dose_FBP"),
    test_set=load_test_set("PWM-LDCT", split="test",
                           task="lung_nodule_5mm"),
    signal_ratio=0.25, modality="CT",
    task=Task("lung_nodule_5mm", metric="auc",
              ground_truth_protocol="majority-vote-2-rads"),
    subpopulation="adult_chest_pwm_l3_test_v1",
    epsilon=0.02, alpha=0.05,
    formulation="aggregate", estimator="bca",
    bootstrap=10_000, seed=42,
)
```

Step 2 — inspect the resulting credential JSON.

```
{
  "schema_version": "pwm-signal-equivalence/v0.1",
  "l2_framework_hash": "sha256:<computed-at-runtime>",
  "credential": {
```

```

171     "method": {"name": "my_method", "code_hash": "sha256:..."},
172     "reference_method": {"name": "FBP_full_dose"},
173     "signal_ratio": 0.25, "modality": "CT",
174     "task": {"name": "lung_nodule_5mm", "metric": "auc",
175             "ground_truth": "majority-vote-2-rads"},
176     "subpopulation": "adult_chest_pwm_l3_test_v1",
177     "epsilon": 0.02, "alpha": 0.05,
178     "estimator": "bca", "bootstrap": 10000, "seed": 42,
179     "verdict": "<PASS|FAIL|INDETERMINATE>",
180     "delta_mean": <reported>,
181     "delta_ci_95": [<lower>, <upper>],
182     "n_test": <reported>
183   }
184 }

```

185 *Step 3 — third-party verification.* A reader given only the credential JSON pulls the framework
186 definition document from the on-chain registry by hash, pulls the test set by the published L3
187 spec hash, runs the same library against `my_method`’s published runbundle, and checks that the
188 recomputed verdict + CI match the published values within $\leq$ Monte-Carlo error of the bootstrap.
189 This is not a hypothetical workflow: it is the acceptance test for every submission to the PWM
190 Low-Dose CT Challenge leaderboard (to be confirmed, 2028).

191 3 Discussion

192 The signal-equivalence framework converts vendor marketing claims into reviewable scientific
193 propositions. A claim of the form “50% dose reduction” is unfalsifiable; a claim of the form “signal-
194 equivalent at ($r = 0.5, T = \text{lung-nodule-AUC}, \varepsilon = 0.02, \alpha = 0.05, \Pi = \text{adult-chest-LIDC}$) against
195 $\mathcal{M}_{\text{ref}} = \text{FBP-at-full-dose}$ ” is falsifiable: re-run the paired-bootstrap on the same Π and the same T ,
196 and the CI either passes or fails the equivalence band.

197 Three implications follow.

198 **Reviewer practice.** A methods paper that does not report a 5-tuple credential, or that reports
199 one with ε or α left unspecified, is not making a comparable claim. Journals adopting the framework
200 as a reporting standard — analogous to CONSORT for randomized controlled trials (Schulz et al.,
201 2010) — would force authors to make claims that subsequent researchers can verify.

202 **Regulatory practice.** The FDA’s draft guidance on AI/ML-based software as a medical de-
203 vice (U.S. Food and Drug Administration, 2024) requires subgroup-stratified performance report-
204 ing; a 5-tuple credential per subgroup operationalizes that requirement. A method certified at
205 ($r, T, \varepsilon, \alpha, \Pi_{\text{adult}}$) but not at $\Pi_{\text{pediatric}}$ has explicit evidence of its subpopulation boundary, which
206 is what regulatory reviewers ask for and rarely receive in current submissions. Because the cre-
207 dential is paired with a content-addressed framework hash, it also satisfies the FDA’s emerging
208 predetermined-change-control-plan (PCCP) expectations (U.S. Food and Drug Administration,
209 2024) for clear delineation of which performance claims are tied to which model version: a creden-
210 tial at `pwm_l2_spec:dose_equivalence_v1:1.0.0` cannot be silently re-evaluated against a later
211 framework version without producing a new credential under the new hash.

Agent practice. As AI agents increasingly query published evidence (Anthropic, 2024), they require evidence that is structured, machine-readable, and verifiable. A 5-tuple credential satisfies all three properties; a free-text claim “the method is equivalent at 50% dose” satisfies none. Anchoring credential schemas to a public registry (Section 4.6) makes the verification step cryptographic rather than reputational.

Adoption pathway. Reporting standards in medical imaging have historically diffused through three channels in sequence: a methods paper that defines the standard; one or two flagship journals that require it for new submissions in a specific subfield; and an FDA or international-regulator citation that closes the loop. We propose the following 36-month adoption trajectory for signal-equivalence credentials.

- **Months 0–12.** Open-source library at v1.0.0 on PyPI; tutorial notebooks for three modalities; three to five external research groups have computed credentials for their own methods. The PWM Low-Dose CT Challenge (to be confirmed, 2028) accepts only submissions with credentials, providing a concrete worked-example community.
- **Months 12–24.** One reconstruction-methods journal (target: *Medical Image Analysis* or *IEEE TMI*) adopts the framework as a recommended reporting standard for low-signal acquisition method papers, parallel to how CONSORT was first adopted by single journals before spreading.
- **Months 24–36.** The FDA cites the framework as one acceptable form of evidence for subgroup-stratified performance reporting under its AI/ML SaMD guidance (U.S. Food and Drug Administration, 2024), complementing rather than replacing existing reader-study evidence. A separate non-imaging methodological community (e.g., pharmacokinetic modeling) considers adopting the same framework for cross-method comparisons under their own modality-equivalent of T_r .

This adoption trajectory is aspirational and contingent on the framework’s empirical performance and the open-source library’s maturation. We commit to honest reporting of adoption progress through the recurring annual challenge review (to be confirmed, 2028).

Failure modes of the framework

The framework can return an actionable PASS or FAIL only when the data and the task are well-specified. We list the failure modes explicitly so users do not over-interpret a verdict in regimes where the framework is silent.

INDETERMINATE verdicts. When the bootstrap CI straddles the equivalence band $(-\varepsilon, \varepsilon)$, the verdict is INDETERMINATE: the data are consistent with both equivalence and non-equivalence at the chosen ε and α . We refuse to force a binary verdict in this regime; honest reporting of INDETERMINATE is part of the framework, not a defect. INDETERMINATE typically indicates one of: (i) insufficient n for the chosen (ε, α) — mitigation: increase n per the sample-size formula; (ii) high per-patient performance variance relative to the margin — mitigation: tighten the subpopulation Π to reduce variance, or relax ε ; (iii) genuinely close-call performance where the method is neither demonstrably equivalent nor demonstrably worse.

Out-of-scope claim types. The framework deliberately does *not* produce credentials for claims of the form:

- *Superiority* (“ M is better than $\mathcal{M}_{\text{ref}}$ at r ”). The framework tests equivalence, not superiority; a separate one-sided test is needed for superiority claims.
- *Cross-modality transfer* (“ M trained on CT works on MRI”). The framework’s $\mathcal{M}$ is single-modality; transfer between modalities requires re-evaluation against the new modality’s T_r .
- *Composition* (“If $M \equiv \mathcal{M}_{\text{ref}}$ at r and $\mathcal{M}_{\text{ref}} \equiv \mathcal{M}'_{\text{ref}}$ at r' , then $M \equiv \mathcal{M}'_{\text{ref}}$ at rr' ”). We argue composition should be re-evaluated rather than derived.
- *Multi-task simultaneity* (“ M is equivalent for tasks T_1 and T_2 ”). A v1 credential is task-specific; v2 will offer a task-bundle extension.

Estimator failure modes. The paired bootstrap is consistent for finite-second-moment performance differences, but: (i) AUC near 1.0 with small n is anti-conservative under the percentile CI — the library defaults to BCa in this regime with an explicit warning; (ii) heavy-tailed metric distributions can require larger B than the default 10,000 — the library reports a Monte-Carlo standard error so users can detect this; (iii) ground-truth label noise that exceeds ε makes any equivalence verdict meaningless — the library refuses to issue a credential when the documented adjudication-protocol disagreement rate exceeds $\varepsilon/2$.

Subpopulation failure modes. A credential under subpopulation Π does not transfer to $\Pi' \neq \Pi$, full stop. Users routinely want to claim transfer; the framework refuses to support that claim and the library produces no aggregate credential across subpopulations. Users wanting a cross-subpopulation guarantee must compute and report separate per-subpopulation credentials.

Aggregate-vs-per-patient. The aggregate formulation (Definition 1) matches clinical-trial endpoint reporting but is silent about per-patient regret. For settings where the cost of a single missed lesion dominates (rare disease, oncology screening, pediatric), the per-patient strengthening (Definition 2) should be opted into. The library exposes both formulations through a single `formulation=` argument; the default is `aggregate`.

Reproducibility caveats. A published credential is reproducible only if the test set, the method, the reference method, the bootstrap seed, and the framework version are all reproducible. The PWM Registry’s content-addressed L2 / L3 specs (Section 4.6) cover the framework and the test set; the method’s code-hash field covers the method. The library refuses to compute a credential when the method or reference method has not been wrapped in a hash-addressable artifact (Docker image with SHA-256 digest, or pinned PyPI release).

4 Methods

4.1 The framework, formally

An imaging modality is a triple $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \mathcal{Y})$ of measurable spaces (patient space, acquisition space, reconstruction space) together with:

- A *reference acquisition operator* $S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{S})$. For each patient $x \in \mathcal{X}$, $S_{\text{ref}}(x)$ is the distribution over raw measurements that the full-signal scanner produces on x (probabilistic because acquisition is noisy).
- A family of *signal-reduction operators* $\{T_r\}_{r \in (0,1]}$ with $T_r : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$. The interpretation: if $s \sim S_{\text{ref}}(x)$, then $s' \sim T_r(S_{\text{ref}}(x))$ is the raw measurement at reference-level scaled by r .

The modality-specific instantiations are given in Table 5.

Modality	T_r instantiation
CT	Poisson-thinning: if $s_{\text{ref}} \sim \text{Poisson}(\lambda(x))$, then $s_{\text{red}} \sim \text{Poisson}(r \cdot \lambda(x))$
MRI	k -space subsampling: $s_{\text{red}} = \Omega_r \cdot s_{\text{ref}}$ with $ \Omega_r / \Omega_{\text{full}} = r$
PET	Poisson-thinning of list-mode counts at rate r
Optical	Reduced exposure time: $s_{\text{red}} \sim \text{Poisson}(\phi(x) \cdot r \cdot t_{\text{ref}})$

Table 5: Modality-specific signal-reduction operators. CT and PET share the Poisson-thinning model; MRI’s T_r is itself stochastic via the mask distribution.

What “modality-general” means, precisely. The cross-modality claim has both a mathematical and an empirical component, and we are careful to distinguish them.

Mathematically, the credential definition (Definition 1) is parametric in $(\mathcal{M}, T_r, T, P)$. Any imaging modality with a well-defined reference acquisition operator S_{ref} and a measurable family of signal-reduction operators $\{T_r\}_{r \in (0,1]}$ admits a credential under this definition. The paired-bootstrap estimator (Algorithm 1) is similarly parametric: it requires only that $P(T(\cdot), g(\cdot))$ produce a real-valued score per case. Both pieces are modality-agnostic by construction; no modality-specific code is required to compile a new modality.

Empirically, the three modalities we ship as validated have been checked against published baselines and clinically meaningful tasks. Other modalities admit credentials of the same form but have not been validated by us:

Modality	T_r status	Validation status
CT (tube-current reduction)	validated	Phase 1 pilot + multi-vendor PWM-LDCT cohort
MRI (k-space undersampling)	validated	fastMRI knee, variable-density Cartesian mask
PET (activity reduction)	validated	NEMA NU-2 IQ phantom, multiple activity levels
Optical / fluorescence	specified	not validated; user-implementable
OCT (A-scan undersampling)	specified	not validated; user-implementable
Ultrasound (channel sparsity)	not specified	open research question
Diffusion-weighted MRI	special case	requires per-direction T_r design
Spectral / dual-energy CT	special case	requires per-energy T_r design

Table 6: Status of each modality under the framework. “Validated” = we ship a worked example. “Specified” = T_r is given in Table 5 but no empirical case-study is included. “Special case” = T_r requires per-axis design beyond the scalar r .

What would falsify the modality-general claim. If the framework’s verdicts on MRI or PET disagreed systematically with established clinical-equivalence assessments at the same operating point,

the modality-general claim would fail. We do not observe such disagreement in our worked examples (Section 2), but call out the falsifiability explicitly because reviewer scrutiny of cross-modality claims is appropriate: any single-modality TOST is uncontroversial, but uniform applicability across physical-acquisition models is the contribution that needs evidence.

A *reconstruction method* is a measurable map $M : \mathcal{S} \rightarrow \mathcal{Y}$. The composed map $M \circ T_r \circ S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$ is “reconstruct from a reduced-signal scan of patient x .” We compare two such composed maps: $\mathcal{M}_{\text{red}} := M \circ T_r \circ S_{\text{ref}}$ (the method under evaluation, on reduced-signal data) and $\mathcal{M}_{\text{base}} \circ S_{\text{ref}}$ (a named reference method on full-signal data; we write $\mathcal{M}_{\text{ref}}$ for shorthand).

A *task* is a measurable map $T : \mathcal{Y} \rightarrow \mathcal{D}$ from reconstructions to a clinical decision space $\mathcal{D}$ (binary detection, bounding boxes, per-voxel segmentation masks, scalar quantification, etc.). Ground truth $g(x) \in \mathcal{D}$ is established by a documented adjudication protocol (multiple board-certified radiologists, majority vote, biopsy confirmation), and the protocol is part of the credential.

A *performance metric* is a functional $P : \mathcal{D} \times \mathcal{D} \rightarrow \mathbb{R}$ evaluating “how well the task output on a reconstruction agrees with ground truth.”

A *subpopulation* is a probability measure Π on $\mathcal{X}$ together with an operational description (recruitment / inclusion criteria a trial coordinator could use to verify draws from Π).

Definition 1 (Signal-equivalence, aggregate formulation). *A reconstruction method M is signal-equivalent to a reference method $\mathcal{M}_{\text{ref}}$ at level $(r, T, \varepsilon, \alpha)$ over subpopulation Π , written $M \equiv_{(r, T, \varepsilon, \alpha, \Pi)} \mathcal{M}_{\text{ref}}$, iff*

$$\left| \mathbb{E}_{x \sim \Pi} \left[\mathbb{E}_{y \sim \mathcal{M}_{\text{red}}(x)} P(T(y), g(x)) \right] - \mathbb{E}_{x \sim \Pi} \left[\mathbb{E}_{y' \sim \mathcal{M}_{\text{ref}}(x)} P(T(y'), g(x)) \right] \right| < \varepsilon \quad (1)$$

with confidence $\geq 1 - \alpha$ under the paired-bootstrap estimator of Section 4.3.

Definition 1 reads: “the population-mean task performance of M on reduced-signal scans is within ε of the population-mean task performance of $\mathcal{M}_{\text{ref}}$ on full-signal scans, with confidence $1 - \alpha$.”

We also define a strictly-stronger *per-patient* formulation:

Definition 2 (Signal-equivalence, per-patient formulation). *A method M is per-patient signal-equivalent to $\mathcal{M}_{\text{ref}}$ at level $(r, T, \varepsilon, \alpha)$ over Π iff*

$$\mathbb{P}_{x \sim \Pi} \left[\left| \mathbb{E}_{y \sim \mathcal{M}_{\text{red}}(x)} P(T(y), g(x)) - \mathbb{E}_{y' \sim \mathcal{M}_{\text{ref}}(x)} P(T(y'), g(x)) \right| < \varepsilon \right] \geq 1 - \alpha. \quad (2)$$

We adopt Definition 1 as canonical because it matches clinical-trial endpoint reporting. Definition 2 is an opt-in strengthening (e.g., for pediatric or rare-disease subpopulations where per-patient regret dominates).

4.2 Type signatures: six clarifications

A naive informal version of Equation (1) conflates six distinctions that must be separated before any theorem can be cleanly stated. We list them here because reviewers may otherwise read the framework as a re-statement of TOST.

(C1) Patient vs image. T_r acts on the *acquisition* of a patient, not on the patient. The patient $x \in \mathcal{X}$ is latent; the framework never observes x directly.

(C2) T_r is an operator, not a scalar multiplier. Writing “ $S_{\text{red}} = r \cdot S_{\text{ref}}$ ” is dimensionally wrong: S_{red} is an operator-valued map, parameterized by r .

(C3) Reference acquisition $\neq$ reference algorithm. “Reference” is ambiguous in the informal version. We separate S_{ref} (the operator, full-signal scan) from $\mathcal{M}_{\text{ref}}$ (the reconstruction method).

(C4) Performance is a function of $(T(y), g(x))$, not (M, S, T) . The metric evaluates the task output against ground truth; the method, scan, and task generate the output but are not its arguments.

(C5) Subpopulation is a measure, not a label. Two subpopulations with the same label but different operational definitions are different subpopulations.

(C6) Aggregate vs per-patient. Definitions 1 and 2 are not equivalent; per-patient implies aggregate, but not conversely.

4.3 Estimator

Given a test set $\{x_1, \dots, x_n\}$ drawn i.i.d. from Π with ground truth $\{g(x_k)\}$, and methods $M, \mathcal{M}_{\text{ref}}$, we estimate Equation (1) via a paired bootstrap on patients (Algorithm 1).

Algorithm 1 Paired-bootstrap signal-equivalence credential

Require: Test set $\{x_k\}_{k=1}^n$ from Π ; ground truth $\{g(x_k)\}$; methods $M, \mathcal{M}_{\text{ref}}$; reduction operator T_r ; task T ; metric P ; margin ε ; significance α ; replicates B ; seed.

```

1: for  $k = 1, \dots, n$  do
2:    $s_{\text{red}} \sim T_r(S_{\text{ref}}(x_k)); \quad a_k \leftarrow P(T(M(s_{\text{red}})), g(x_k))$ 
3:    $s_{\text{ref}} \sim S_{\text{ref}}(x_k); \quad b_k \leftarrow P(T(\mathcal{M}_{\text{ref}}(s_{\text{ref}})), g(x_k))$ 
4: end for
5: for  $b = 1, \dots, B$  do
6:   Sample  $n$  indices  $\{i_1, \dots, i_n\}$  with replacement from  $\{1, \dots, n\}$ 
7:    $\Delta_b \leftarrow \frac{1}{n} \sum_{j=1}^n (a_{i_j} - b_{i_j})$ 
8: end for
9:  $[\ell, u] \leftarrow$  central  $(1 - \alpha)$  percentile interval of  $\{\Delta_b\}$ 
10: return verdict  $\leftarrow$  if  $[\ell, u] \subset (-\varepsilon, \varepsilon)$  then PASS; else if  $[\ell, u] \cap (-\varepsilon, \varepsilon) = \emptyset$  then FAIL;
    else INDETERMINATE

```

We use $B = 10,000$ as the default. Sampling pairs (rather than the two arms independently) is essential because a_k and b_k are correlated through the patient x_k .

Sample-size formula. For a target (ε, α) and a per-patient performance-difference standard deviation σ with pairing correlation ρ , the leading-order normal-approximation bound is

$$n \geq \left(\frac{z_{1-\alpha/2}}{\varepsilon} \right)^2 \sigma^2 (1 - \rho). \quad (3)$$

For AUC-type metrics with $\sigma \approx 0.10$ and $\rho \approx 0.5$, the canonical target $(\varepsilon = 0.02, \alpha = 0.05)$ yields $n \geq 192$. We pre-register $n \geq 200$ patients per credential as the field default. A finite-sample Bernstein correction is documented in Supplementary Section [\[TODO: supp section\]](#).

Coverage of percentile CIs near boundary. For AUC near 1.0 the standard percentile bootstrap CI is anti-conservative. We provide a BCa option in `pwm_dose_equivalence` and recommend BCa for any credential with reported AUC > 0.95 on test sets with $n < 200$. The library prints a warning when this regime is detected.

4.4 Open-source library

The `pwm_dose_equivalence` library (Apache 2.0, https://pypi.org/project/pwm_dose_equivalence/) implements Definition 1 for CT, MRI, and PET. The user-facing API is a single function call:

```
cred = signal_equivalence_credential(  
    method          = M,                      # callable: scan -> image  
    reference_method = Mref,                   # callable: scan -> image  
    test_set        = data,                    # iterable of (patient, gt)  
    signal_ratio     = r,                      # in (0, 1]  
    modality         = "CT" | "MRI" | "PET",  
    task             = Task(name, metric, target),  
    subpopulation    = "operational_description",  
    epsilon          = 0.02,  
    alpha            = 0.05,  
    formulation      = "aggregate" | "per_patient",  
    estimator        = "percentile" | "bca" | "delong",  
    bootstrap        = 10_000,  
    seed             = 42,  
)
```

Returns a `SignalEquivalenceCredential` object with the verdict (PASS / FAIL / INDETERMINATE), the bootstrap CI, the sample-size sanity check, and a JSON serialization conforming to the schema in Section 4.6.

4.5 Software engineering rigor

The library is intended to be a long-lived community tool, not a paper-companion artifact that lapses six months after publication. We invested in three classes of engineering quality consistent with that intent.

Testing. The library ships with [TODO: N_{tests}] unit tests and [TODO: N_{integ}] integration tests covering: the paired-bootstrap estimator against an analytical-normal reference solution; BCa-vs-percentile-vs-DeLong agreement at controlled σ and ρ ; sample-size formula vs simulation; cross-modality API call-signature equivalence; JSON-credential round-trip and L2 hash stability; and explicit failure-mode behavior (empty test set, mismatched ground-truth shape, NaN propagation, deterministic seed handling). Continuous integration runs on every commit against Python 3.10, 3.11, and 3.12 on Linux, macOS, and Windows runners; the target test-coverage floor is [TODO: $\geq 90\%$] measured by `coverage.py`.

Type discipline. The codebase is fully typed (PEP 484 annotations enforced by `mypy -strict`), with explicit `TypedDict` schemas for credential JSON inputs and outputs. The choice of strict typing is deliberate: when the library is used as a reporting standard, silently-coerced argument types are a class of error that produces credentials that look valid but are not. Strict types surface the error at the API boundary.

Documentation and tutorial notebooks. The library’s documentation site (<https://pwm-dose-equivalence.readthedocs.io/>) hosts API reference (auto-generated from docstrings), three tutorial notebooks (one per validated modality), and a “Credential reading guide” for reviewers and regulators who need to interpret published credentials without writing code themselves. The notebooks are tested in CI to ensure they remain runnable as the API evolves.

Versioning policy. Releases follow semantic versioning. MAJOR bumps break the credential JSON schema; MINOR bumps add modalities, estimators, or task types without breaking existing credentials; PATCH bumps are bug fixes that preserve credential reproducibility. The current version is **[TODO: 0.1.0]**; we target 1.0.0 at journal acceptance with a five-year support commitment.

Release cadence and governance. The library is maintained by the PWM Protocol Foundation with a documented contribution policy at the repository. Pull requests touching the framework definition, the estimator, or the JSON schema require sign-off from two of three named maintainers; pull requests touching documentation, tests, or worked examples require one. The contribution policy is published as `CONTRIBUTING.md` in the repository.

4.6 Cryptographic anchoring to public registry

The credential schema is content-addressed and registered as an L2 specification on the PWM blockchain registry. A published credential identifies the framework version by hash:

```
{
  "schema_version": "pwm-signal-equivalence/v0.1",
  "l2_framework_hash": "sha256:...",
  "credential": { ... }
}
```

The hash uniquely identifies the framework definition the credential commits to. A reader verifies the credential by (i) pulling the framework definition document by hash from the registry, (ii) re-running the paired bootstrap on the same test set, (iii) checking that the recomputed verdict matches the published verdict. This anchoring eliminates a class of benchmark-drift attacks where a method’s reported performance is silently re-evaluated against a different framework version after submission.

5 Data availability

The three modality validators in this paper use publicly available datasets, all of which require registration but are free for academic use:

- **CT.** The multi-vendor PWM-LDCT dataset ([to be confirmed, 2027a](#)) on PhysioNet credentialled access; supplementary CT data from LIDC-IDRI ([Armato III et al., 2011](#)) (NBIA registration) and the AAPM 2016 Low-Dose CT Grand Challenge ([AAPM, 2016](#)) (Mayo Clinic request).
- **MRI.** The fastMRI knee dataset ([Zbontar et al., 2020](#)) (NYU registration), variable-density Cartesian undersampling subset.

- **PET.** NEMA NU-2 IQ phantom acquisitions at four activity levels (full, 50%, 25%, 10%), acquired in-house and released alongside this paper on PhysioNet.

The exact test-split manifest hashes used for every credential in this paper are recorded as on-chain L3 specifications and listed in Supplementary Table [TODO: supp tab]. A reader who pulls the manifest by hash from the PWM registry is guaranteed bit-identical content to what we evaluated.

6 Code availability

The signal-equivalence library, `pwm_dose_equivalence`, is released under Apache 2.0 at https://pypi.org/project/pwm_dose_equivalence/ (pip-installable); source repository at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework/pwm_dose_equivalence/; documentation at <https://pwm-dose-equivalence.readthedocs.io/>. The exact library version used for every credential in this paper is [TODO: pinned-version-at-submission]. All scripts, configuration files, tutorial notebooks, and Docker images required to reproduce each table and figure are at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework. The CI pipeline that runs every reproducibility test on every commit is at https://github.com/integritynoble/low_dose_CT/actions.

Reporting Summary

A Nature Methods Reporting Summary — using the format current at submission, covering subgroup-stratified performance reporting per (U.S. Food and Drug Administration, 2024), code-and-data-availability transparency, statistical disclosure, and ethics — accompanies this manuscript as Supplementary Information. The summary cross-references the relevant Methods subsections so reviewers can locate each item in the main text.

Author contributions

[TODO: Fill at submission per CRediT taxonomy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.]

Competing interests

[TODO: Declare at submission. The PWM Protocol Foundation receives no direct revenue from credential issuance; foundation-affiliated authors hold no financial interest in specific reconstruction methods evaluated under the framework.]

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